A General Deep Learning Framework Guided by Sparse Matching for Disparity Estimation in High-Resolution Satellite Stereo Imagery

Guangbin Zhang, Yonghua Jiang, Jingyin Wang, Yunming Wang, Shaodong Wei, Bin Du, and Meilin Tan

***Abstract*— In the field of photogrammetry and remote sensing, the task of satellite stereo image disparity estimation (SSIDE) has long been recognized as both challenging and important. Currently, deep-learning methods are gaining prominence in the SSIDE domain. However, the inconsistency between stereo images and ground truth makes the fine training and accurate inference of SSIDE networks extremely difficult. Furthermore, the existence of textureless and repeated texture areas in satellite images complicates the execution of end-to-end SSIDE networks, especially in areas with variable illumination conditions. In this study, a sparse matching point-guided disparity estimation (SMP-DE) general framework was introduced to address such concerns. SMP-DE employed sparse matching point-guided data evaluation and distillation (SMP-DED) for fault-tolerant training and ensuring unbiased guidance training as well as reliable reasoning. In addition, SMP-DE executed optimization for the disparity estimation network across various feature spaces by integrating sparse matching point-guided feature contrastive registration (SMP-FCR) and matching cost uniqueness constraint (MCUC) modules. Therefore, SMP-DE can mine homogenous features and model low-entropy matching costs in challenging regions. Experimental results demonstrated that SMP-DE has outstanding disparity estimation accuracy and generalization compared with other advanced methods. Furthermore, the proposed SMP-DED exhibited excellent flexibility and generality, since it can be combined with various disparity estimating networks, giving the networks an accuracy boost on a range of datasets. In summary, SMP-DE provides a novel perspective for end-to-end SSIDE research.**

***Index Terms*— Data evaluation and distillation, disparity estimation, multiple optimization spaces, satellite stereo images, universal deep learning framework.**

I. Introduction

IN THE fields of photogrammetry and remote sensing, disparity estimation (dense stereo matching) of stereo image pairs has consistently posed a challenge [1], [2]. Disparity estimation can produce dense corresponding pixels and precise disparity values for epipolar rectified stereo image pairs, enabling the inference of 3-D information from the images. This technology is essential for rapid and large-scale 3-D reconstruction of ground scenes from satellite images. Accurate satellite stereo image disparity estimation (SSIDE) guarantees smooth progress in ground-based 3-D reconstruction projects. It holds significant application value for fields, such as digital Earth, virtual reality, and 3-D smart cities [3], [4], [5].

After a long period of research and development, disparity estimation methods have undergone continuous improvement and optimization, which has gradually enriched their technical systems. Disparity estimation methods currently in use can be broadly divided into two categories: traditional manual feature-based methods and learning-based methods. The matching process for traditional solutions can be roughly divided into several key stages: matching cost computation, cost aggregation, disparity calculation, and disparity refinement. Graph cuts [6], belief propagation [7], semiglobal matching (SGM) [8], and patch matching [9] are examples of classical methods. Although these methods have led to significant improvements in the field of disparity estimation, they remain limited to areas with abrupt light changes and textureless images. Furthermore, different from natural scenes in computer vision, these methods face more problems when dealing with intricate large-scale satellite images [10], [11].

With the advancement of computer technology and the expansion of annotated data, data-driven learning-based methods have experienced rapid growth in visual semantics [12]. Earlier researchers have realized the potential of learning-based methods for feature extraction and representation learning. However, at the time, the robustness of these methods and the performance of the computer hardware limited their practicality in the field of disparity estimation. In response to these challenges, researchers have integrated learning-based methods in a single pipeline of classical disparity estimation methods to enhance performance and usability [1], [13]. For example, learning-based methods have been employed in classical approaches to replace the original matching cost [14], [15], [16], disparity optimization [17], [18], and disparity refinement [19], [20] steps. This early integration strategy successfully enhanced traditional methods and led to the establishment of a solid foundation for the

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subsequent advancement of end-to-end disparity estimation techniques.

Based on the constant advancement of deep learning technology and the continued addition of sample data [21], [22], [23], [24], the feasibility of learning-based deep neural networks for end-to-end reasoning in geometry has been confirmed [22], [25], [26]. This significantly accelerated the development of deep learning for disparity estimation. Numerous outstanding end-to-end networks have emerged as the dominant players in the research community focused on disparity estimation [21], [22], [23], [24]. Representative models, such as the 2-D architecture DispNet [22], 3-D architecture GC-Net [25], and transformer architecture STTR [27], have made significant advancements. Remarkable network models, such as EdgeStereo [28], PSM-Net [26], CSTR [29], and ELFNet [30], have also introduced novel technological concepts into this field. End-to-end disparity estimation methods have led to significant improvements in both accuracy and efficiency, further demonstrating their potential applications in this field.

The end-to-end deep learning disparity estimation paradigm has attracted considerable interest in the field of photogrammetry. To help deep learning networks better adapt to remote sensing landscapes, researchers have built and enhanced datasets for SSIDE over the past few years, such as urban semantic 3-D (US3D) [31], [32], and WHU-stereo [33]. Simultaneously, academics have explored the application of end-to-end disparity estimation methods to satellite imagery. Qin et al. [34] utilized the concept of the complementary sharing of low-level features and achieved preliminary success by employing U-Net and PSM-Net for segmentation and disparity estimation in satellite imagery. Building on the same principle, Rao et al. [35] designed BGA-Net to integrate semantic segmentation and disparity estimation tasks within a single network by relying on multitask feature sharing to achieve improved disparity estimation. He et al. [36] used multiscale features to construct a pyramid-shaped cost volume and supervised the regression of features at each level for accurate disparity estimation.

The work discussed above provides preliminary evidence for the efficacy of deep learning-based SSIDE methods. However, end-to-end methods still have to be improved before they can produce accurate SSIDE. The following is a summary of identified challenges and deficiencies.

1) Existing learning-based disparity estimation methods rely heavily on finely annotated ground truth for regression. However, they do not consider the prevalence of data errors. In particular, in the fields of photogrammetry and remote sensing, stereo images and disparity ground truth inaccuracies in satellite images are common. Two primary causes have been identified: first, the temporal gaps between light detection and ranging (LiDAR) point clouds and stereo images are frequently ignored. When building the disparity ground truth, LiDAR data collecting timeframes frequently differ from image collection times, and natural evolution and human activities lead ground objects to alter in unexpected ways over time (see Fig. 1). Second, errors are introduced during ground truth creation, such as grid interpolation and point-cloud fusion [37]. These factors have the effect of making stereo-image pairs and disparities noncorresponding. The accuracy and generality of learning-based methods for SSIDE are significantly affected by errors in the data and lead to misguided results. It makes ensuring the consistency of training data in SSIDE a crucial challenge.
2) Challenging regions represented by textureless and texture-repetitive zones are more common in satellite stereo images. Moreover, the changes in lighting conditions have caused the stereo images to have variations in shadow area and reflection strength, which has caused challenging regions to become more complex. Therefore, existing learning-based methods have often produced blurry or even incorrect results in these regions. The reason is that the networks only use disparity regression with simple geometric relation encoding between training. Thus, developing diversified optimization networks to move beyond dependence on single disparity regression and establishing reliable dense feature matching in challenging areas of satellite stereo images are urgent issues that need to be addressed.

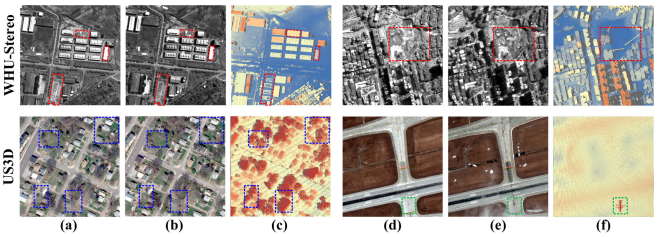


Fig. 1. Inconsistencies between stereo images and ground truth. (a) and (d) Left-view satellite images. (b) and (e) Right-view satellite images. (c) and (f) Ground truth. The rectangular frames highlight areas where discrepancies between the images and ground truth exist. The red frames indicate inconsistencies caused by human construction activities, the blue frames show inconsistencies due to vegetation growth, and the green frames point to inconsistencies caused by ground object movement.

To effectively address these issues, a general sparse matching point-guided disparity estimation (SMP-DE) deep learning framework was introduced in this study. This framework comprises a sparse matching point-guided data evaluation and distillation (SMP-DED) training strategy and a multiple feature space optimization (MFSO) network. SMP-DED conducts a precise quality assessment and encoding of data pairs in disparity estimation datasets. Based on the quality assessment, SMP-DED implements a coarse-to-fine fault-tolerant training strategy and provides unbiased guidance during training. Furthermore, it can be easily extended to other types of deep learning disparity estimation networks and datasets. During model optimization, MFSO integrates sparse matching point-guided feature contrastive registration (SMP-FCR) and matching cost uniqueness constraint (MCUC) modules. SMP-FCR uses sparse matching prior information to encourage the infinite proximity between feature vectors of corresponding points in the feature space. In addition, the MCUC analyzes and models the entropy of disparity estimation to quantify

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ambiguity. It employs entropy optimization to regularize the matching cost and reduce uncertainty in disparity estimation. The SMP-FCR and MCUC modules effectively standardize the feature representation space and matching cost space, thereby significantly reducing the matching noise and instability in challenging areas of satellite imagery. The main contributions of this study are as follows.

1) A general SMP-DE framework is proposed. The technical processes and modules of the framework significantly increase the accuracy of disparity estimation. Moreover, the carefully designed training strategy can be effectively extended to other disparity estimate networks or datasets.
2) The data evaluation and coarse-to-fine distillation training strategy within the framework facilitate rational and refined network training for existing satellite disparity estimation datasets. In addition, the multiple-feature space optimization network significantly enhances the disparity estimation accuracy and generalization.
3) Thorough ablation studies and universality tests of the framework were conducted, in addition to large-scale comparative experiments. The results validate the effectiveness and generality of the technical strategies and components proposed in this work.

The remainder of the manuscript is organized as follows. Relevant works for disparity estimation are reviewed in Section II. The proposed disparity estimation framework is described in Section III. Comparative experiments and analyses are provided in Section IV. The methods employed in this study are discussed in Section V. Finally, a comprehensive summary of this study is presented in Section VI.

II. Related Works

End-to-end inference disparity estimation networks have been continuously improved and refined. In this section, a brief overview of the development of mainstream end-to-end disparity estimation networks in computer vision and remote sensing areas is provided.

A. End-to-End Disparity Estimation Method in Computer Vision

Because of the excellent performance of end-to-end deep-learning methods in extracting features, researchers have introduced them into the field of disparity estimation to improve the performance of algorithms. Existing end-to-end disparity estimation networks can generally be categorized into networks based on 2-D architectures, 3-D architectures, and transformer architectures based on their feature-encoding type and geometry construction strategies.

DispNet [22] is a milestone network that introduced end-to-end methods to the field of disparity estimation. It uses an encoder–decoder-type 2-D convolutional architecture to extract features from stereo images and compute similarity, thereby inferring disparity. HD^3 [38] adopts coarse-to-fine pyramid feature extraction and a feature-warping strategy in its encoder. It also sets up a dense embedding space in its decoder to estimate the disparity. In addition, HD^3 combines discrete distribution estimation to determine the uncertainty of the disparity estimation. EdgeStereo [28], [39] simultaneously learns the common tasks of disparity estimation and edge segmentation in its network framework, effectively enhancing depth discontinuity areas in the disparity results. Although these 2-D architecture-based networks have successfully introduced end-to-end methods into disparity estimation tasks, strict and explicit geometry encoding has not been established [1].

To establish geometric matching properties, GC-Net [25] constructs a 4-D cost volume based on 3-D feature vectors and uses 3-D convolution for further feature matching. Finally, GC-Net employs a soft-argmin function to process features and obtain the disparity. Following the same 3-D feature-matching design paradigm, PSMNet [26] uses spatial pyramid pooling layers for multilevel feature extraction and construction, effectively improving the accuracy of the disparity estimation. GA-Net [40] emphasizes the benefits of cost aggregation by designing local-guided and semi-global aggregation layers to model local and global costs, respectively. Therefore, GA-Net efficiently ensures the disparity estimation accuracy. StereoNet [41] aims at reducing the computational and memory requirements of 3-D architectures by limiting the 3-D encoder–decoder process to low resolution. Refined disparity estimation results of StereoNet were obtained by up-sampling and refining the coarse disparity. GWC-Net [42] constructs a new 3-D network cost aggregation mode by combining feature concatenation and vector correlation for cost-volume construction. The cost aggregation mode offers superior feature representation and more accurate disparity estimation. Three-dimensional architecture-based disparity estimation methods effectively avoid black-box disparity estimation through explicit geometry reasoning and enhance disparity estimation accuracy. However, these methods require a preset disparity search range and struggle to capture features over long distances.

Integrated with transformer architectures, STTR [27] is initially designed with alternating self-attention and cross-attention to model the correlation of features from stereo views. By combining the transformer, STTR can perform disparity estimation from the pixel sequences. CSTR [29], which is based on the STTR paradigm, further added a context enhancement path to strengthen the capture of remote and global semantics. ELFNet [30] combines transformer and 3-D convolution architectures and performs intraevidential fusion based on multilevel predictions and interevidential fusion between various disparity estimation methods. ELFNet can effectively predict disparity estimation and output uncertainty. Transformer-based networks significantly integrate global information and do not require the construction of a fixed disparity search range. However, the computational complexity of this type of network increases significantly.

B. End-to-End Disparity Estimation Method in Remote Sensing

In recent years, end-to-end deep-learning methods have been used in the field of SSIDE. Bidir-EPNet [43] enhances PSM-Net and adapts it for disparity estimation tasks in

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stereo satellite image pairs. BGA-Net [11] is based on the idea of a multitask-guided architecture. It designs a unified main module, semantic segmentation module, feature matching module, bidirectional guidance module, and bidirectional fusion module to simultaneously execute disparity estimation and semantic segmentation tasks. The two tasks mutually benefit from enhanced feature sharing and complementary result fusion optimization through a bidirectional attention mechanism, thereby boosting performance. HMSM-Net [36] constructs a pyramid-type cost volume, performs robust context feature matching at feature levels with different receptive fields, and uses a hierarchical cost-fusion approach for adaptive cross-scale cost-volume fusion and output. S^2Net [44] incorporates disparity results into classification feature maps and uses semantic features to enrich the cost volume. Meanwhile, S^2Net uses the duality of stereo image pairs to augment the data and obtain better results. The aforementioned end-to-end networks have achieved good disparity estimation on various satellite images.

In summary, in the disparity estimation domain of satellite scenes, end-to-end deep-learning methods have greatly simplified traditional algorithms and achieved significant accuracy improvements. However, satellite imagery contains extensive areas with large-scale lighting changes, shadow occlusions, repetitive textures, and nontextures, presenting a significant challenge to existing disparity estimation methods. In addition, there may be significant inconsistencies between the constructed ground truth and corresponding stereo-image content. The performance of end-to-end networks is highly susceptible to effects when trained on data with inconsistencies. These issues must be considered and addressed to realize practical applications.

III. Proposed Method

A. Framework Overview

The SMP-DE framework is guided by prior sparse matching information to facilitate the implementation of a coarse-to-fine fault-tolerant training strategy based on data evaluation and distillation. Meanwhile, sparse matching information promotes the design of an optimized network in the SMP-DE framework across multiple feature spaces. The main architecture of SMP-DE is shown in Fig. 2.

B. SMP-DED Training Strategy

In current SSIDE datasets, a temporal discrepancy is notable between stereo satellite image pairs and LiDAR data used to construct the ground truth [32], [33]. Temporal differences in data increase the difficulty of maintaining strict content consistency. In particular, when considering the temporal evolution of ground objects, the spatiotemporal mismatch between various data sources is further exacerbated. The presence of incongruence errors between the ground truth and imagery can negatively affect the training of models that use stereo image pairs as inputs and ground truth as an optimization target. Therefore, effectively assessing and representing the quality of each data pair in a dataset is crucial to facilitate rational and refined training.

To overcome the aforementioned problems, we investigated sparse matching point-guided data quality evaluation-encoding and coarse-to-fine fault-tolerance training strategy.

1) Sparse Matching Point-Guided Quality Evaluation and Encoding of the Disparity Estimation Dataset: In this study, a procedural and robust data evaluation method that combines existing sparse matching algorithms was introduced to effectively quantify and describe data errors. The overall execution process is illustrated in Fig. 3. The disparity estimation dataset $O_m = \{L_i, R_i, D_i\}^n_{i,n\in\mathbb{R}}$ consists of n data pairs, each consisting of left image L, right image R, and the disparity ground truth D. For each data pair, sparse matching points $P_L = \{x^j_L, y^j_L\}^m_{j,m\in\mathbb{R}}$ and $P_R = \{x^j_R, y^j_R\}^m_{j,m\in\mathbb{R}}$ were obtained using sparse points detection and matching algorithms, such as SIFT [45], SURF [46], and FLANN [47]. However, because of the presence of large areas without texture, repetitive textures, and varying lighting conditions in satellite imagery, the robustness of the feature extraction and matching results of sparse matching algorithms is reduced. Therefore, in this study, a ratio purification method was employed for initial filtering of the sparse matching results. Benefiting from the epipolar rectification characteristics of stereo images, an epipolar constraint was employed in this study to refine the matching points. The detailed process for obtaining, examining, and filtering sparse matching points is outlined in Algorithm 1.

In the algorithm, SparseDetec() refers to the sparse feature point detection algorithm, P represents initially obtained sparse feature points, Desc denotes the feature descriptor of the sparse matching points, MatchDesc() is the feature point matching algorithm, MP represents the initial set of matching points obtained, α is the initial threshold for the ratio purification method, β is the minimum required number of valid sparse matching points, γ is the threshold increment for the ratio purification method when the minimum number of sparse matching points is not met, P_m is the set of sparse matching points preliminarily optimized using the ratio purification method, d represents the Euclidean distance of feature points, append() represents the function of adding elements to the set, P_L is the left matching point in the matching set P_m, P_R is the right matching point in the matching set

Algorithm 1 Implementation of Obtaining and Filtering Sparse Matching Points

Input: O_m Data pair in the SSIDE dataset $O_m = \{L, R, D\}$
Output: $\hat{P}_m$ Obtained optimal sparse matching points $\hat{P}_m = \{\hat{P}_L, \hat{P}_R\}$

$P_L, Desc_L = SparseDetec(L)$
$P_R, Desc_R = SparseDetec\ (R)$
$MP = MatchDesc(desc_L, desc_R)$
$P_m, \hat{P}_m = [\ ], [\ ]$
while *true*:
  for MP_L, MP_R ***in*** MP:
    if $MP_L \cdot d < (\alpha * MP_R \cdot d)$:
      $P_m \cdot append(MP_L, MP_R)$
  if $len(P_m) >= \beta$:
    break
  else:
    $P_m = [\ ]$
    $\alpha = \alpha = \gamma$
  for P_L, P_R ***in*** P_m:
    if $P_L \cdot y - P_R \cdot y < \delta$:
      $\hat{P}_m \cdot append(P_L, P_R)$

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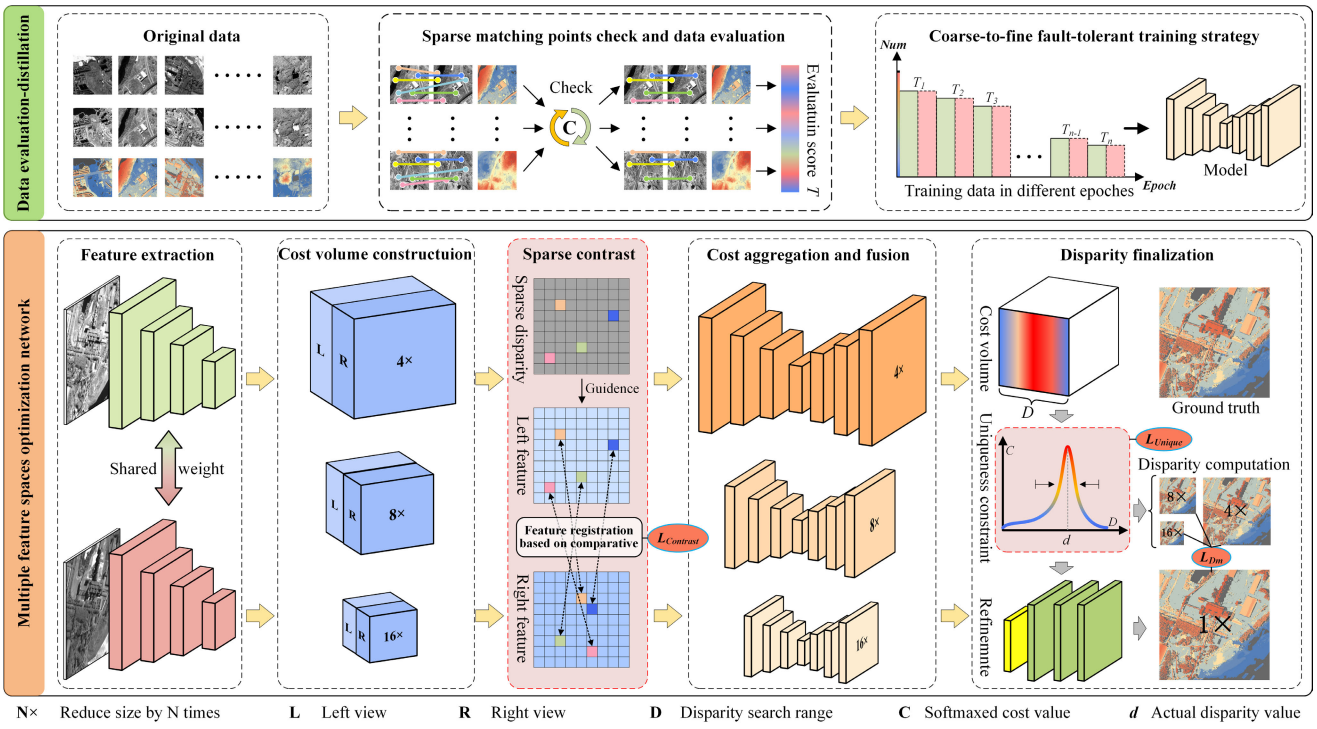


Fig. 2. Overall technical architecture of the SMP-DE general framework.

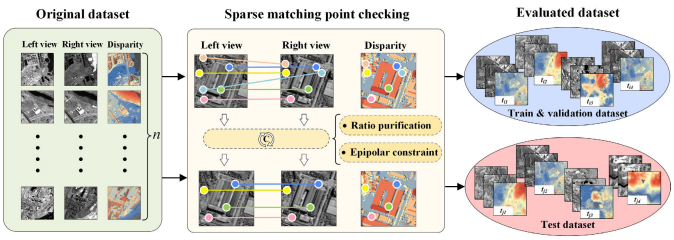


Fig. 3. Flowchart for disparity estimation data pair evaluation and encoding guided by sparse matching points.

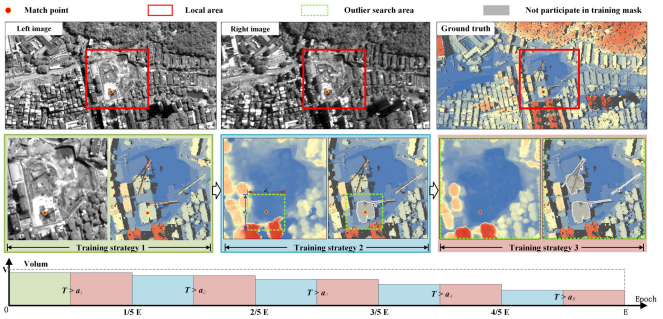


Fig. 4. Sparse matching point-guided coarse-to-fine fault-tolerance training strategy.

P_m, Py represents the ordinate of the matching point, and δ is the elevation difference threshold used for filtering sparse matching points based on the epipolar characteristics of stereo images. After a series of checks and filters, the set of optimized matching points $\hat{P}_m$ with $\hat{m}$ matching points is obtained, which includes the corresponding left matching points $\hat{P}_L = \{x_L^j, y_L^j\}_{j,\hat{m}\in\mathbb{R}}^{\hat{m}}$ and right matching points $\hat{P}_R = \{x_R^j, y_R^j\}_{j,\hat{m}\in\mathbb{R}}^{\hat{m}}$. The corresponding sparse disparity $S = \{s^j\}_{j,\hat{m}\in\mathbb{R}}^{\hat{m}}$ is calculated using the following equation:

$$s^j = x_L^j - x_R^j. \tag{1}$$

The quality assessment score T for each data pair in the dataset can be calculated based on the mean absolute difference between the sparse disparity values and corresponding ground-truth disparity values

$$T = \frac{\sum_{j=1}^{\hat{m}} \left| s^j - d^j \right|}{\hat{m}}. \tag{2}$$

The calculated assessment score T was assigned to the corresponding data pair to indicate the reliability, as shown in (3). In addition, the optimized sparse disparity values S are encoded for use in Section III-C

$$O_m = \{L_i, R_i, D_i, T_i, , S_i\}_{i,n\in\mathbb{R}}^{n}. \tag{3}$$

2) Sparse Matching Point-Guided Coarse-to-Fine Fault-Tolerance Training Strategy: Diverse types of data can provide networks with rich feature representations. Furthermore, a sufficient quantity of data helps the network better capture the distribution patterns of the data, thereby reducing the risk of overfitting. In summary, ample and varied data enhance the robustness and generalization of the network, ensuring that it performs well across scenes under varying conditions. However, relying solely on data-pair assessment scores to split the dataset and exclude erroneous data can significantly reduce the data diversity. Ultimately, the reduction in the diversity of data may lead to challenges in providing strong and comprehensive feature representation for networks.

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To utilize the existing SSIDE data both reasonably and effectively, a coarse-to-fine disparity estimation fault-tolerant training strategy was designed in this study (see Fig. 4). This strategy ensures that the model can make an accurate and robust fit using a large amount of data with varying quality. During the training process, the data quality assessment scores T were closely integrated and served as key indicators to determine whether a data pair should be included in the training. In the initial stage of training, a lenient data assessment score threshold φ_i was set. This threshold ensured that a large amount of data could be included in the training while excluding data with notable errors. This stage provided a rich data foundation for model training, helping it to learn more comprehensive and abundant feature representations. As the model gradually converged through learning, the data assessment score threshold became increasingly stringent and gradually excluded inaccurate training data. The data involved in training at each stage were defined using the following equation:

$$O_m^k = \{L_i, R_i, D_i, T_i, S_i\}_{i,n\in\mathbb{R}}^n, \quad T_i < \varphi_k \tag{4}$$

where O_m^k represents the dataset at the kth training stage. Progressive and dynamic training strategies helped the network capture the overall data distribution characteristics and more general features during the initial training stages. In the later stages of training, retaining the refined data helped the model learn the real data distribution and capture the refined feature representation, thereby enhancing the training quality.

Although the transition from a large amount of coarse training data to a smaller amount of refined training data effectively enhances the network performance, inconsistencies may exist in both data pairs with poor- and better-quality assessments. Therefore, in this study, three refined training strategies were constructed by closely integrating the encoded sparse disparity S. These strategies were designed to avoid guidance feedback from error-prone data regions during training. In the first training strategy, all data involved in the training were fully supervised

$$L_{\text{dm1}} = \text{Smooth}L_1(\text{pred}, D) \tag{5}$$

where Smooth$L1()$ denotes the smooth $L1$ loss function and pred denotes the disparity prediction result of the network. In the second training strategy, encoded sparse matching points were used as anchors to determine if the difference between ground truth is in the confidence interval $[-\varepsilon, \varepsilon]$. If the difference did not exceed the confidence interval, the fully supervised method in (5) was applied. If the difference exceeded the threshold ε, further determination was carried out in the neighborhood. Neighborhood regions in which the difference between the disparity estimation results and ground truth also exceeded the threshold ε were set as the excluding region mask_2. The loss function for this training strategy is as follows:

$$\begin{aligned}\text{mask}_2 &= [S(\eta,1,1) - D(\eta,1,1) > \varepsilon] \\ &\quad \times \left[\text{pred}\left(\eta, d_x, d_y\right) - D\left(\eta, d_x, d_y\right) > \varepsilon\right] \end{aligned} \tag{6}$$

$$L_{\text{dm2}} = \text{Smooth}L_1(\text{pred}\times(1-\text{mask}_2), D\times(1-\text{mask}_2)) \tag{7}$$

where (η, d_x, d_y) refers to a local area centered around the matching point η, with a length d_x and width d_y. The third training strategy was entirely based on self-training. The entire disparity estimation results of the network were compared with the ground truth. When the difference exceeded the confidence interval $[-\varepsilon, \varepsilon]$, an excluding region mask_3 was established. The loss function is defined based on the following region:

$$\text{mask}_3 = (\text{pred} - D) > \varepsilon \tag{8}$$

$$L_{\text{dm3}} = \text{Smooth}L_1(\text{pred}\times(1-\text{mask}_3), D\times(1-\text{mask}_3)). \tag{9}$$

The SMP-DED training strategy is illustrated in Fig. 4. It allows for a more refined exclusion of ground-truth error areas that might mislead the network, thereby enhancing the learning performance and fitting effectiveness.

C. MFSO Network

Although the SMP-DED training strategy can effectively address errors in the training data for SSIDE, existing disparity estimation networks struggle to adequately handle challenging regions in stereo satellite images. Previous SSIDE networks relied solely on disparity regression for training to achieve mapping from stereo images to disparity results. However, the training strategy is overly simplistic and direct, making it challenging to effectively address common issues in textureless and repeating texture areas as well as areas with illumination and shadow variation.

In this study, we employed the MFSO to address the aforementioned challenges. As shown in Fig. 2, the overall architecture of the MFSO can be summarized as follows: in the feature extraction stage, stereo images are first processed using a feature-extraction architecture with shared weights. Subsequently, extracted multilevel features are included in a multilevel pyramid cost volume. Then, the produced features are subjected to cross-scale feature extraction and aggregation by employing serial 3-D convolution. Ultimately, the multilevel cost volumes are subjected to disparity computations and high-resolution disparity refinements in the matching cost space. It is worth noting that, MFSO also utilized SMP-FCR and MCUC to promote network optimization from multiple perspectives. SMP-FCR conducted contrastive learning within the left and right feature spaces. It effectively facilitates the capture of homogenized features for corresponding points and heterogeneous features for noncorresponding points in the feature space. MCUC executed disparity estimation uniqueness constraint. It promotes the optimal solution for disparity estimation in the matching cost space. The combination of two innovative technologies in MFSO can effectively improve the accuracy of disparity estimation in challenging areas.

1) Sparse Matching Point-Guided Feature Contrastive Registration: The SMP-FCR scheme was constructed (see Fig. 5). The SMP-FCR was designed to effectively build a strong correlation model for sparse features in stereo image pair feature spaces. It aims to mine the homogeneous features of corresponding points and encourage the network to capture

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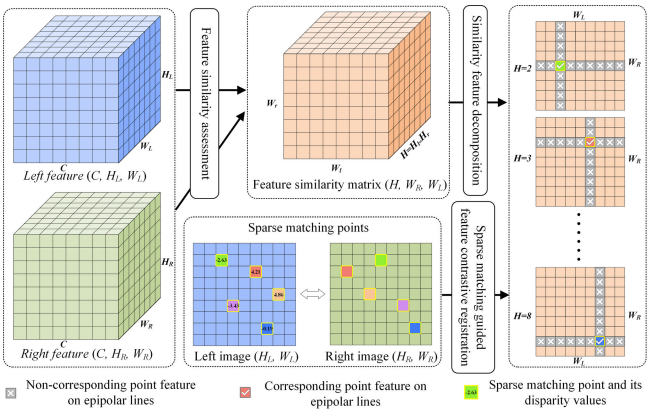


Fig. 5. Module architecture of SMP-FCR.

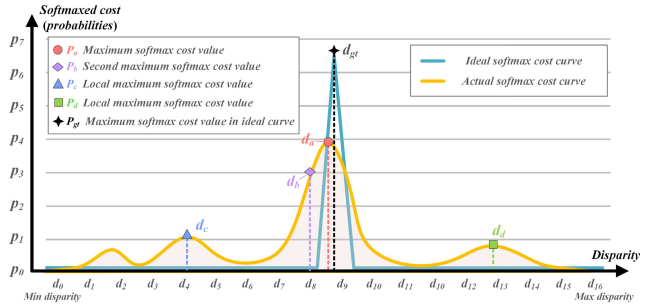


Fig. 6. Ideal and actual distribution of matching cost volume curves in the disparity estimation network.

differentiated features of noncorresponding points. Specifically, the left feature F_L and right feature F_R captured by the feature-extraction architecture were used to construct a similarity matrix M along the channel dimension

$$M_{(H\times W_L\times W_R)} = e^{\frac{F_{L(H_L\times W_L\times C)}\cdot F_{R(C\times H_R\times W_R)}}{\left\|F_{L(H_L\times W_L\times C)}\right\|\times\left\|F_{R(C\times H_R\times W_R)}\right\|\times\tau}} \tag{10}$$

where C, H, and W represent the number of feature channels, height, and width, respectively, and τ is the temperature of the similarity matrix. Furthermore, we decomposed the feature similarity matrix along the H dimension into feature similarity subvectors. Based on the encoded sparse matching points S, we determined the corresponding feature subvectors M_{sim} and noncorresponding feature subvectors $M_{\text{non-sim}}$

$$M_{\text{sim}} = \left\{M\left(w_L^j, w_R^j\right)\right\}_{j,\hat{m}\in\mathbb{R}}^{\hat{m}} \tag{11}$$

$$\begin{aligned} M_{\text{non-sim}} = \Big\{ & M\left(\rho, w_R^j\right), M\left(w_L^j, \sigma\right) \Big| \\ & \rho \in \left\{1, 2, \ldots, w_L^j - 1, w_L^j + 1, \ldots, W_L\right\}, \\ & \sigma \in \left\{1, 2, \ldots, w_R^j - 1, w_R^j + 1, \ldots, W_R\right\}\Big\}_{j,\hat{m}\in\mathbb{R}}^{\hat{m}} \end{aligned} \tag{12}$$

where w_L^j and w_R^j are the horizontal coordinates of corresponding points in sparse matching points S, $M(w_L^j, w_R^j)$ are the feature subvectors in the coordinate positions of (w_L^j, w_R^j) of similarity matrix M, and W_L and W_R are the sizes of the feature similarity matrix M. Based on the obtained corresponding and noncorresponding point feature subvectors, a contrastive learning loss is constructed

$$L_{\text{Contrast}} = \frac{\log(1 + M_{\text{non-sim}})}{\hat{m}\times(W_L - 1) + \hat{m}\times(W_R - 1)} - \frac{\log(1 + M_{\text{sim}})}{\hat{m}}. \tag{13}$$

Contrastive registration learning based on sparse matching points can be used to effectively narrow the distance between corresponding points in the feature space. Meanwhile, the module pushes away the feature distance of the noncorresponding points along corresponding epipolar lines. As a result, SMP-FCR encourages the network to learn more distinctive features in textureless areas and areas with lighting condition changes and enhances the feature-matching accuracy.

2) Uniqueness Constraint on Matching Cost (MCUC): Satellite imagery often contains a multitude of objects with nearly identical visual characteristics and arranged in a regular pattern. These repetitive texture areas can significantly affect disparity estimation methods, causing their matching cost curves to exhibit a multimodal distribution (as shown in Fig. 6). This distribution can result in blurring or incorrect matching. To address this issue, we introduced the MCUC, which was integrated into our network. The MCUC was specifically designed to tackle the multimodal distribution problem in dense matching cost curves and aims to generate precise and unique matching results.

In SMP-DE, a disparity calculation is conducted using soft argmin. Specifically, multilevel cost volumes C are transformed into a continuous disparity expectation P using the softmax function. Subsequently, a probabilistic weighted sum operation is employed to obtain the subpixel-level disparity prediction, as follows:

$$\text{pred} = \sum_{i=D_{\min}}^{D_{\max}} i \times P_i = \sum_{i=D_{\min}}^{D_{\max}} i \times \sigma(-C) \tag{14}$$

where σ is the softmax function, $D_{\max}$ and $D_{\min}$ are the maximum and minimum values of the disparity prediction range, respectively. The final disparity prediction result pred is the weighted sum of the candidate disparity values i and their disparity expectations P_i. The ideal disparity probability distribution generally reaches its peak near the true disparity and exhibits unimodal uniqueness (blue curve in Fig. 6). However, the actual disparity expectation is represented by the yellow curve in Fig. 6. The results of previous studies showed that disparity estimation errors are positively correlated with the degree of multimodal distribution of matching cost curves [48]. Therefore, in the SMP-DE, the area and peak ratio of the expected disparity curve are used to construct a disparity uniqueness constraint loss to ensure the accuracy of the output disparity

$$L_{\text{Unique}} = \left(1 - \sum_{i=D_{\min}}^{D_{\max}} P_i\right) + \mu \times \log\frac{(P_a + P_b) + (P_c + P_d)}{|(P_a + P_b) - (P_c + P_d)|} \tag{15}$$

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where P_a is the probability value at the peak of the disparity prediction probability curve, P_b is the second largest value of the probability curve for disparity, P_c and P_d are the maximum values of the local regions of the disparity prediction probability curve, and μ is the weight of the peak ratio portion.

The SMP-DE with the MCUC module can guide network optimization from another perspective and promote dense disparity estimation to generate unique and accurate disparity estimation values. Thus, the MCUC effectively reduces the fuzzy and erroneous disparity estimation in repetitive texture regions.

IV. Experiment Results

In this section, we provide an overview of the experimental data, data division standards, and evaluation metrics. Specific details related to the experiment, including the training environment, training parameters, and implementation details of the comparative methods, are described. Finally, we provide a comprehensive comparison of the accuracy and complexity of the methods.

A. Datasets and Evaluation Metrics

In the experiments conducted in this study, we employed two representative public SSIDE datasets, WHU_Stereo [33] and US3D [32], to conduct sequential training using our method and other state-of-the-art (SOTA) networks. The performance of these methods was subjected to a comprehensive evaluation.

WHU_Stereo is a challenging SSIDE dataset. High-resolution stereo satellite image pairs in the dataset were acquired using a dual-line-array stereoscopic camera on the Gaofen-7 satellite. The camera provided near-synchronous stereo satellite images with a spatial resolution exceeding 0.8 m. The ground truth geometry in the dataset was constructed by processing the point clouds obtained from airborne LiDAR data. The data construction process included four steps to ensure high-quality epipolar images and accurate disparity values: point cloud alignment, epipolar image generation, point cloud projection, occlusion removal, and data benchmark generation. Ultimately, after the aforementioned series of processes, the dataset consisted of 1757 data pairs covering six cities: Shaoguan, Kunming, Yingde, Qichun, Wuhan, and Hengyang.

US3D is a large-scale disparity estimation dataset with accompanying semantic annotations. The dataset contains WorldView-3 satellite stereo images that have been epipolar-corrected as well as geometric labels constructed based on aerial LiDAR data. The ground sampling distance of the image data was approximately 35 cm, acquired by satellites over different periods covering Jacksonville and Omaha between 2014 and 2016. LiDAR point-cloud data were used to construct the ground truth. In addition, the dataset included semantic labels corresponding to the left-view images for conducting related semantic tasks. The disparity estimation dataset of US3D encompasses a total of 4292 pairs of stereo images and ground truth, making it one of the largest disparity estimation datasets currently available.

TABLE I
Division of Training, Validation, and Testing Data for WHU_Stereo and US3D Datasets in the Experiment

Dataset	City	kind	Training	Validation	Testing
WHU-Stereo	Qichun	Original	600	280	280
		Selected	600	27	27
	Other 5 Cities	Original	/	/	877
		Selected	/	/	334
US3D	Omaha	Original	1500	653	653
		Selected	1500	155	155
	Jackson	Original	/	/	2139
		Selected	/	/	927

Before conducting comparative experiments using different methods, we divided the WHU_Stereo and US3D datasets. To ensure the generalizability assessment of the methods used in the experiment, WHU_Stereo and US3D were divided into two subsets based on the geographical characteristics of the images (as shown in Table I). In the WHU_Stereo dataset, we designated data from Qichun as the training data (original data) and data from the other five cities as the generalizability assessment data (extended data). For the US3D dataset, data from Omaha served as training data (original data) and data from the Jackson area served as generalizability assessment data (extended data). Notably, both the validation and test sets were selected based on the data evaluation results ($T < 1.0$) to ensure the reliability of the method performance assessments. In the original data, the validation and the test subsets used the same data.

To quantitatively describe the reliability of the methods, we employed the average endpoint error (EPE) and fraction of erroneous pixels ($D1$) metrics. In the experiments, we also used the accuracy variation (Δ) metric based on the extended dataset to quantitatively assess the generalizability of the methods. The EPE, $D1$, and Δ metrics are defined as follows.

EPE: This metric represents the absolute average difference between the predicted disparity and ground-truth disparity in valid areas (lower is better)

$$\mathrm{EPE} = \frac{1}{\kappa}\sum_{i=1}^{\kappa}\left|\mathrm{pred}_i - D_i\right| \tag{16}$$

where k is the number of valid grid cells in the ground truth, pred_i is the disparity predicted using the method for the ith grid cell, and D_i is the disparity of the ground truth for the ith grid cell.

D1: This metric represents the proportion of the number of absolute disparity errors within a threshold compared with the actual number of valid grid cells (lower is better)

$$D1 = \frac{\sum_{i=1}^{\kappa} I\left(\left|\mathrm{pred}_i - D_i\right| > b\right)}{k} \times 100\% \tag{17}$$

where b is the error threshold and $I(\theta)$ is the Iverson bracket, which takes the value of 1 when the condition θ inside is satisfied and 0 otherwise. Based on past research experience [36], [44], b was set to 3 pixels in this study.

Δ: This metric indicates the changes in the corresponding evaluation indicators in different test sets or methods.

In addition, the complexity of the disparity estimation methods involved in the experiments was evaluated and compared based on the number of parameters, number of forward

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inference floating point operations (Flops), and inference time according to the WHU_Stereo extended test set.

B. Experiment Setup

1) Training Details: All experiments were conducted on a computer equipped with an NVIDIA GeForce RTX 4070 12GB GPU and an Intel[1] Core[2] i5-13600 KF CPU. Traditional and deep learning disparity estimation methods were implemented on Ubuntu 20.04 in combination with OpenCV 3.4.16 and Pytorch 2.0.0.

During the training phase of the experiments, the batch size for all deep learning networks was set to 1 and adaptive half-precision training was employed. For the WHU_Stereo dataset, all deep learning networks were effectively trained for 100 epochs. Given the large scale of the US3D dataset, disparity estimation networks underwent 60 training epochs on this dataset. The Smooth$L1$ loss function was used during training to calculate the feedback losses. Valid disparity estimation ranges for the WHU_Stereo and US3D datasets were set to $[-128, 64]$ and $[-96, 96]$, respectively. All disparity estimation networks based on the 3-D convolutional neural network (CNN) architecture used the Adam optimizer with an initial learning rate of 1e-3 and weight decay of 1e-5 for end-to-end training. For disparity estimation networks based on the 2-D CNN architecture, an Adam gradient optimizer with an initial learning rate of 1e-4 was used. Considering the uniqueness of transformer-based networks, the training strategy for the STTR was maintained as previously described [27]. During the training of STTR, an AdamW optimizer with a weight decay of 1e-4 was used. The learning rate for the feature extractor and transformer structure weights was set to 1e-4, while that for the context adjustment structure was set to 2e-4. The networks used an equidistant learning-rate adjustment strategy during training. In the CNN, the learning rate was halved every 15 and 9 epochs for the WHU_Stereo and US3D datasets, respectively. The learning rate decay epoch and weight of the transformer network were set to 1 and 0.99, respectively. During training, the networks used weights with the minimum loss on the validation set as the benchmark for evaluations of the original and extended test sets.

2) Method Implementation: In the experiments, we conducted a comprehensive comparison of the proposed SMP-DE method with five SOTA methods. These five methods include SGBM, FADNet [49], PSM-Net, STTR, and HMSM-Net, which represent traditional methods, methods based on 2-D architecture, methods based on 3-D architecture, methods based on transformer architecture, and methods specifically designed for satellite imagery, respectively.

1) *SGBM:* The algorithm is an enhancement of the SGM algorithm that combines block matching and the Birchfield–Tomasi metric for cost calculation. It is open source and available in OpenCV. In this experiment, the window size for the SGBM was set to five. The disparity smoothness parameters were set to 600 and 2400. The uniqueness ratio for uniqueness detection was set to 15. The maximum error threshold allowed for left–right consistency checks was 1 and the disparity connectivity condition determination difference was set to 2.
2) *FADNet:* This network does not use 3-D convolutional layers. Instead, it employs stacked 2-D convolutional correlation layers to ensure the efficiency of disparity computation within the network. It features a multiscale output strategy and deep residual structures to facilitate the rapid learning of deep model parameters. The weights for the multiple output loss calculations in FADNet were set to 0.32, 0.16, 0.08, 0.04, 0.02, 0.01, and 0.005, respectively.
3) *PSM-Net:* This network uses multilevel pyramid pooling layers to integrate global context features, followed by stacked hourglass modules composed of a 3-D convolutional CNN for regularization of the cost volume. In the network, the regularized cost volumes at the three levels were calculated for disparity estimation and the weights for losses at these three levels were set to 0.5, 0.7, and 1.0.
4) *STTR:* Because the US3D and WHU_Stereo datasets do not include the ground truth for occluded areas, the binary cross-entropy loss was not used for training the occlusion area prediction in the experiments. In the STTR, six self- and six cross-attention layers were used. The attention stride was set to four and the feature descriptor channel dimension was set to 128. Furthermore, given the characteristics of self- and cross-attention, the STTR does not set a specific disparity search range.
5) *HMSM-Net:* This network includes a pyramid-type feature extraction and fusion process capable of outputting regularized cost volumes at three resolution levels. In addition, HMSM-Net performs upsampling optimization based on multilevel low-resolution cost volumes to produce more refined disparity maps. The loss weights for the three levels of resolution cost volume and refined disparity map output were set to 0.5, 0.7, 1.0, and 0.6.
6) *SMP-DE:* The initial ratio purification threshold α was set to 0.5, the minimum number of effective sparse matching points β was 100, the ratio purification incremental threshold γ was 0.05, and the height difference screening threshold δ was 1.25 pixels. The data evaluation score thresholds for different training stages $(\varphi_1, \varphi_2, \varphi_3, \varphi_4, \varphi_5)$ were set to 5.0, 4.0, 3.0, 2.0, and 1.0 for the WHU_Stereo dataset and 3.4, 2.8, 2.2, 1.6, and 1.0 for the US3D dataset. The negligible error threshold for disparity prediction ε was set to 3 pixels and the length d_x and width d_y of the search area centered around the matching anchor points η were both set to 64 pixels. In addition, the SMP-FCR and MCUC modules will participate in network training at stages where T is less than φ_3. The loss function of the SMP-DE is defined as follows:

$$L = L_{\mathrm{Dm}} + \upsilon L_{\mathrm{Contrast}} + \omega L_{\mathrm{Unique}}. \tag{18}$$

where the weights ν and ω were set to (1, 0.1) for the WHU_Stereo dataset and (0.01, 0.1) for the US3D

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TABLE II

QUANTITATIVE EVALUATION RESULTS OF VARIOUS DISPARITY ESTIMATION METHODS FOR THE WHU_STEREO DATASET

Dataset	Method	Qichun		Other 5 cities		Accuracy decay	
		D1	EPE	D1	EPE	ΔD1	ΔEPE
WHU_Stereo	SGBM	12.51%	4.0404	13.58%	5.3044	**↓1.07%**	↓1.2640
	FADNet	22.30%	2.6177	38.64%	5.7596	↓16.34%	↓3.1419
	STTR	20.66%	2.7454	32.74%	5.1205	↓12.08%	↓2.3751
	PSMNet	22.13%	2.3345	29.31%	3.5252	↓7.18%	↓1.1907
	HMSM-Net	10.74%	1.5984	13.95%	2.1429	↓3.21%	↓0.5445
	SMP-DE	**9.04%**	**1.3633**	**11.59%**	**1.8463**	↓2.55%	**↓0.4830**

* ↓ represents a decrease in evaluation accuracy. Bold and underlined values are optimal.

TABLE III

QUANTITATIVE ASSESSMENT RESULTS FROM VARIOUS METHODS FOR THE US3D DATASET

Dataset	Method	Omaha		Jackson		Accuracy decay	
		D1	EPE	D1	EPE	ΔD1	ΔEPE
US3D	SGBM	8.90%	2.5739	12.07%	3.0184	**↓3.17%**	↓0.4445
	FADNet	7.68%	1.3455	17.49%	2.1154	↓9.81%	↓0.7699
	STTR	4.60%	1.0987	12.93%	1.7982	↓8.33%	↓0.6995
	PSMNet	7.95%	1.4042	15.27%	1.9215	↓7.32%	↓0.5173
	HMSM-Net	4.50%	1.0229	11.26%	1.5088	↓6.76%	↓0.4859
	SMP-DE	**3.75%**	**0.9284**	**9.02%**	**1.3427**	↓5.27%	**↓0.4143**

dataset, respectively. The weight μ for the peak ratio part of the uniqueness constraint loss L_{Unique} was set to 0.01.

C. Experiment Result

All comparative methods and our proposed SMP-DE were trained on the WHU_Stereo and US3D datasets in the same experimental environment. The optimal weights obtained during the training process were then used for disparity prediction and a comprehensive evaluation of both the original and extended test sets.

1) Comparison of WHU_Stereo Dataset: Table II presents a quantitative comparison of the accuracy of the various methods used in the experiment for the Qichun test set and the test sets of the other five cities. The traditional SGBM achieves a low accuracy for the original test set. The EPE of the SGBM was only 4.0404, which is the lowest among the evaluated methods. Its $D1$ metric was 12.51%, which is comparatively better. This indicates that, although the disparity estimation results of the SGBM deviate from the ground truth, they do not produce a large number of outlier estimation results. Furthermore, the accuracy of SGBM for the extended test set did not significantly decline. This demonstrates that traditional methods, represented by the SGBM, exhibit good generalizability in disparity estimation tasks. The reliability of traditional methods does not significantly decrease because of changes in the spectral distribution and geographical attributes of the data.

Compared with the traditional SGBM method, end-to-end methods generally improved the EPE accuracy of disparity estimation on the original Qichun test set. However, FADNet, STTR, and PSMNet showed a significant gap in the $D1$ metric compared with the SGBM. This phenomenon is due to the lower image resolution of the WHU_Stereo dataset, where existing deep-learning methods tend to produce blurry disparity predictions in challenging scenes, leading to a large number of disparity estimation errors. In addition, both FADNet and STTR exhibited significant accuracy degradation in the extended test set. This is attributed to the limited scale of the data in the WHU_Stereo dataset, which challenges the robustness of feature representation in networks with 2-D CNN and transformer architectures. PSMNet and HMSM-Net, which are based on a 3-D CNN architecture, outperformed other types of end-to-end disparity estimation methods in terms of accuracy. Furthermore, the performance decay of these two methods was insignificant for the extended dataset.

Finally, our SMP-DE improved the disparity estimation results on both test sets, achieving the best evaluation scores ($D1$ scores of 9.04% and 11.59% and EPE scores of 1.3633 and 1.8463, respectively). Furthermore, SMP-DE exhibited a very low accuracy degradation on the extended test set. This indicates that the proposed method has good robustness and generalizability for dense disparity estimation.

Fig. 7 shows the disparity prediction results of the various methods for the WHU_Stereo dataset. The SGBM performs accurate disparity inference in simple areas but tends to fail in areas with repetitive textures (black-framed areas), light reflection–shadow changes (red-framed areas), and textureless regions (yellow-framed areas). FADNet, which lacks explicit geometric relationships for inference, produces significant disparity deviations in areas with abrupt spectral changes (e.g., water areas). In addition, the disparity estimation results of FADNet for the extended test set are easily misled by the spectral characteristics of ground objects, leading to numerous anomalies in the disparity estimation. STTR considers too much distant semantic information in geometric relation constructions. Therefore, STTR is easily misleading in complex local scenes, resulting in an incorrect disparity estimation. For the extended test set, STTR produces more fragmented and uneven disparity predictions. PSMNet and HMSM-Net, which use 3-D architectures, produce disparity estimates that closely match the ground truth of the test set. However, these two networks tend to produce sudden disparity changes in textureless areas and fragmented disparity predictions in light reflection-shadow change areas. The SMP-DE proposed in this study produces excellent disparity estimation results in textureless areas, repetitive texture areas, and complex spectrum change areas because it considers feature contrast learning and disparity uniqueness constraints, optimizes the network from multiple perspectives and effectively optimizes the feature matching in these areas. Furthermore, regardless of whether the original Qichun test set or the extended test set was used, SMP-DE achieved disparity estimation results that are the closest to the ground truth.

2) Comparison of US3D Dataset: Table III presents the quantitative evaluation results of various disparity estimation methods for the US3D dataset. Because of the smaller ground sampling distance in US3D, which provides richer and finer details of ground object textures in the imagery, all disparity estimation methods exhibited an improvement in accuracy

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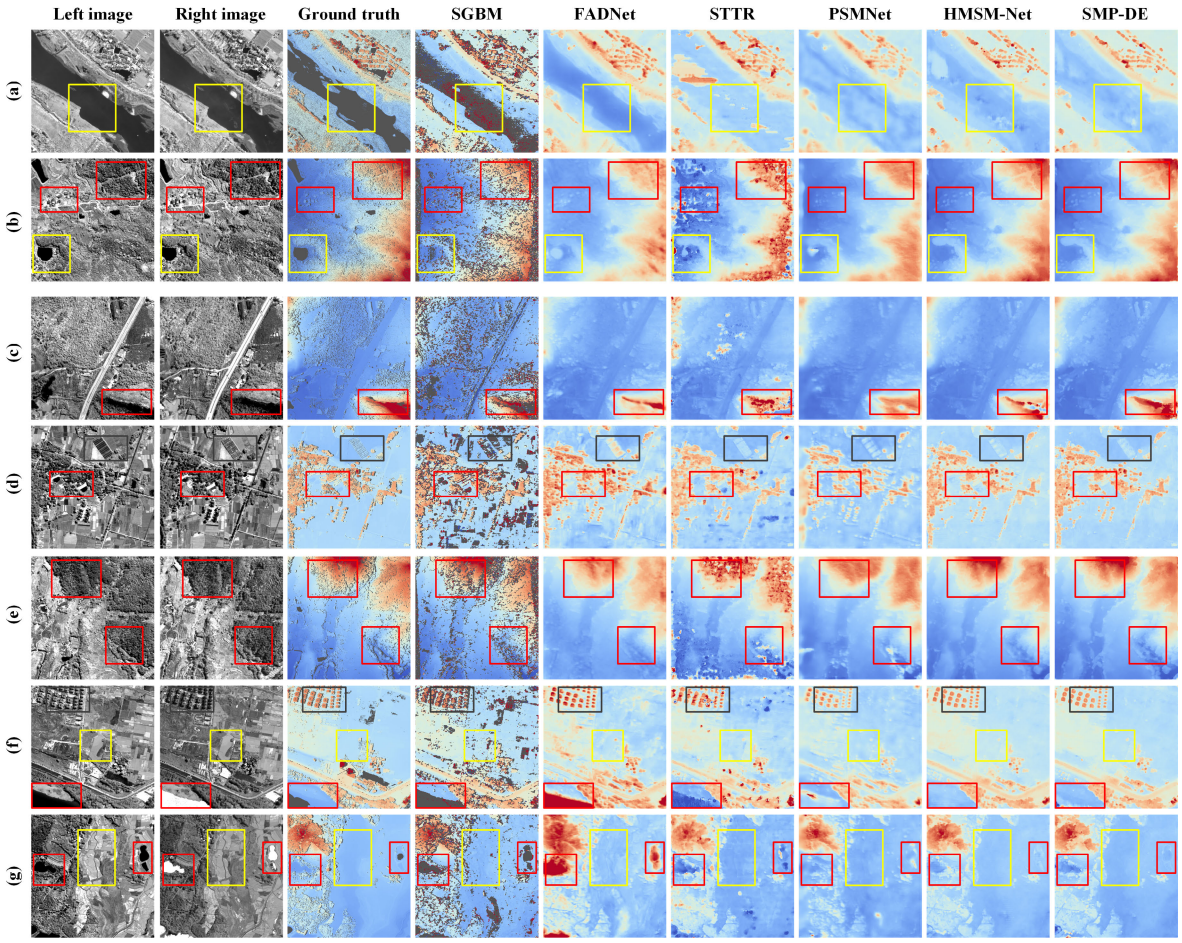


Fig. 7. Visualization of disparity estimation results by various methods for the WHU_Stereo dataset. (a) and (b) Are from the Qichun test set, image numbers are QC_262 and QC_210, respectively. (c)–(g) Are images from the other five cities, with image numbers YD_446, YD_398, YD_334, YD_394, and YD_220, respectively. Areas within the yellow frames are weakly textured and textureless regions; red frames highlight regions of changes in reflection brightness and shadow area caused by variations in the angle of solar illumination; and black frames indicate areas with repetitive textures.

compared with those based on the WHU_Stereo dataset. The STTR method achieves the most significant improvement in accuracy, benefiting from fine local ground information that allows it to effectively consider both local and long-distance semantic features. Good extraction and integration of short- and long-distance semantic relationships enabled STTR to achieve a high disparity estimation accuracy for the Omaha test set. However, the transformer architecture of the STTR requires a large amount of data for robust feature modeling, and the limited training data in Omaha affects its generalization performance on the extended dataset. In the Omaha test set, the SGBM showed the lowest accuracy; however, its excellent generalizability led to minimal accuracy degradation for the extended Jackson test set. FADNet and PSMNet do not have significant advantages in terms of evaluation accuracy or generalizability. HMSM-Net exhibited further improvements in both disparity estimation accuracy and generalizability. The SMP-DE proposed in this study achieved the best disparity estimation accuracy for both the Omaha and Jackson test datasets (EPE of 0.9284 and 1.3427 and $D1$ of 3.75% and 9.02%, respectively). Furthermore, SMP-DE ensured network generalizability across different regions, showing the lowest EPE accuracy degradation of only 0.4243 for the extended Jackson dataset and the second-lowest $D1$ accuracy degradation of 5.27%.

Figs. 8 and 9 show the disparity estimation results of the various methods for the Omaha and Jackson test sets, respectively. In the SGBM, effective matching points were difficult to obtain in areas without texture (blue-framed areas) or in regions with shadow variations (red-framed areas). FADNet produced fragmented and incomplete disparity results for bridges, houses, and trees. Furthermore, the results of FADNet have numerous areas of disparity estimation failure in the Jackson extended test set. The disparity estimation results of STTR were smooth but included erroneous matches in textureless and shadow-variation areas, which are more pronounced in the extended test set. In PSMNet, because of the large proportion of continuous convolutional upsampling, the disparity estimation results tended to be blurry at the edges of small-scale ground objects. HMSM-Net has suboptimal disparity visualization effects; however, its disparity estimation in textureless and repetitive texture areas (green-framed areas) remained unstable. The proposed SMP-DE ensured consistency of disparity in textureless areas, continuous stability of disparity estimation in repetitive texture areas, and smooth disparity edges in shadow-change areas. Furthermore, the

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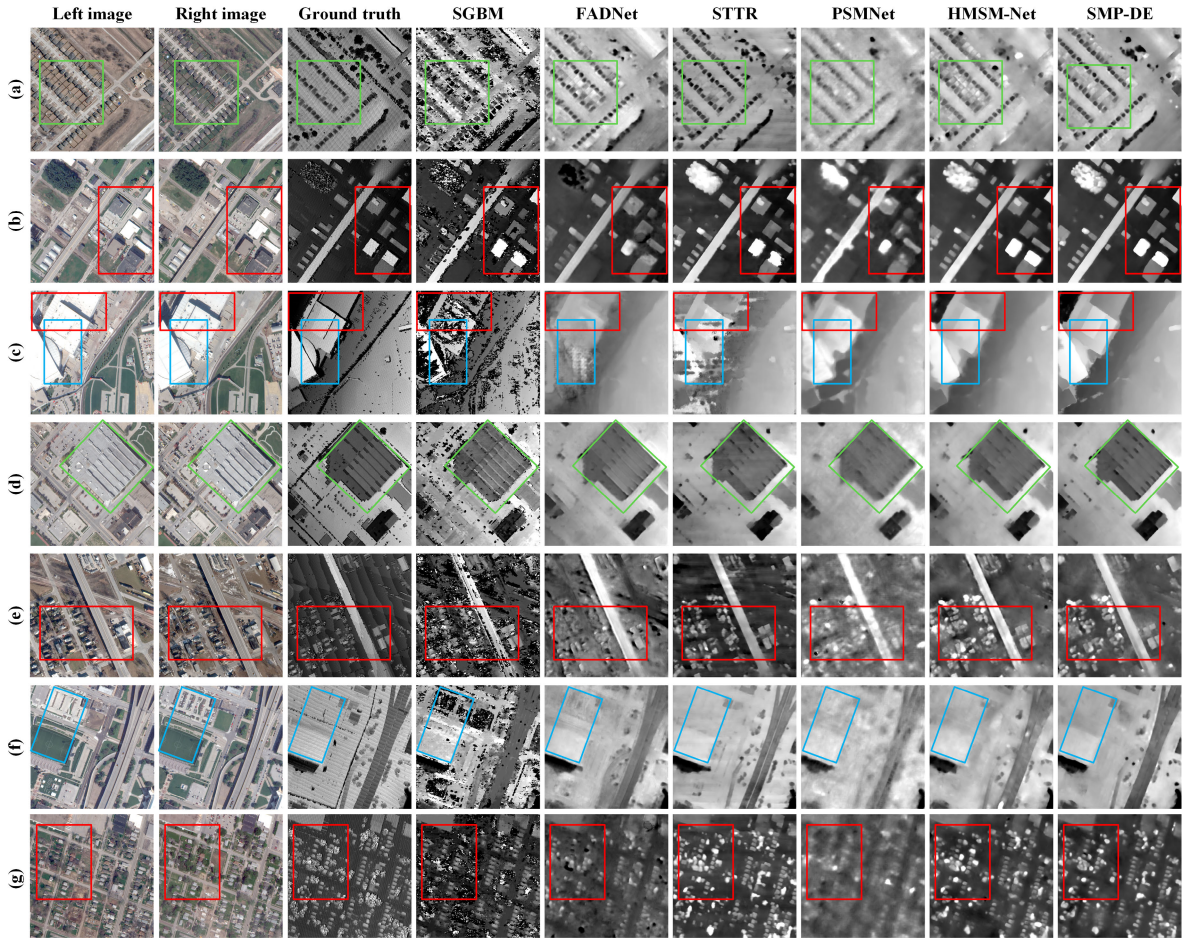


Fig. 8. Visualization of disparity estimation results by various networks for the US3D-Omaha test set. The image numbers for (a)–(g) are OMA391_023_029, OMA315_002_001, OMA251_037_043, OMA244_038_040, OMA353_010_015, OMA247_002_037, and OMA124_029_033, respectively. In these visualizations, green-framed areas represent regions with repetitive textures; red-framed areas are shadow change areas; and blue-framed areas indicate textureless regions.

accuracy of the Jackson extended test set is nearly consistent with that of the original Omaha test set.

D. Complexity Analysis

Table IV displays the quantitative evaluation results of the complexity of each algorithm. The STTR exhibits the longest inference time due to its complex transformer architecture. FADNet, based on the 2-D convolutional architecture, requires a large number of convolution and feature fusion layers to achieve disparity inference capability, resulting in the highest parameter and Flops quantity. HMSM-Net demonstrates the lowest complexity. The SMP-DE proposed in this article also shows strong competitiveness in parameter quantity, computational complexity, and inference time (all second lowest), and thus presents a certain advantage in the execution efficiency of our method.

TABLE IV
QUANTITATIVE EVALUATION OF THE COMPLEXITY OF DISPARITY ESTIMATION METHODS

Method	Parameters (M)	Flops (GFlops)	Inference time (s)
SGBM	/	/	152.91
FADNet	145.006	969.44	98.56
STTR	1.718	851.07	214.83
PSMNet	5.224	233.31	175.63
HMSM-Net	1.309	54.38	76.28
SMP-DE	1.420	56.19	80.64

V. DISCUSSION

In this section, we describe the ablation study conducted to functionally validate the technical components of our SMP-DE and test the generality of the designed SMP-DED training strategies. In addition, we present the verification performed for the robustness of the SMP-DE on extensive test sets and other types of sensor test sets.

A. Ablation Studies

In this experiment, we either removed or added various technical components within SMP-DE to create a series of network variants. These network variants were trained and evaluated using the WHU_Stereo dataset to verify and discuss the effectiveness of each technical component in enhancing the accuracy of disparity estimation. The specific structures of the variant networks are as follows. SMP-DE-V1: A network variant that uses the basic network backbone architecture of SMP-DE. It includes only four parts: feature extraction,

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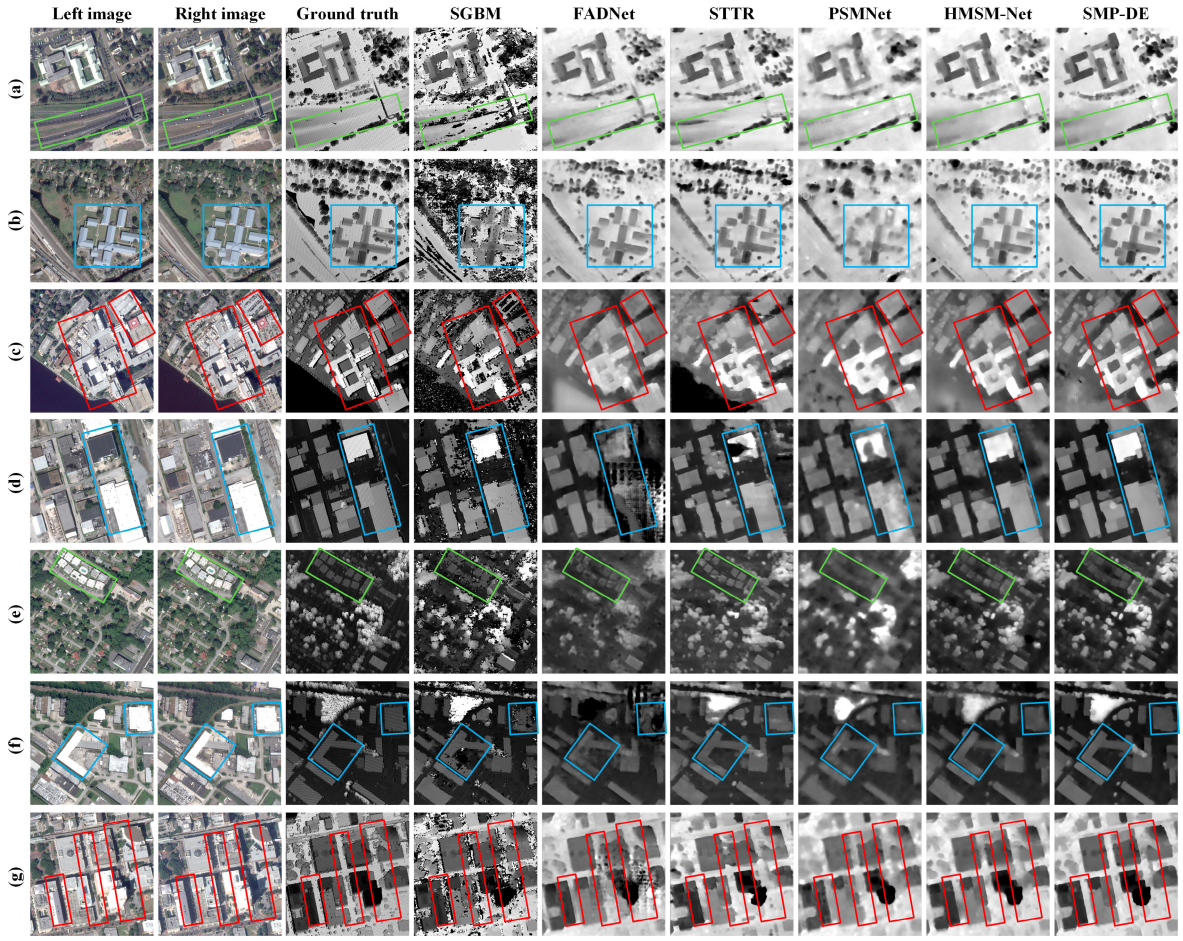


Fig. 9. Visualization of disparity estimation results by various methods for the US3D-Jackson test set. The image numbers for (a)–(g) are JAX_113_011_006, JAX_105_009_023, JAX_251_004_025, JAX_156_002_016, JAX_427_003_023, JAX_467_016_014, and JAX_165_003_005, respectively. In these visualizations, green-framed areas represent regions with repetitive textures; red-framed areas are shadow change areas; and blue-framed areas indicate textureless regions.

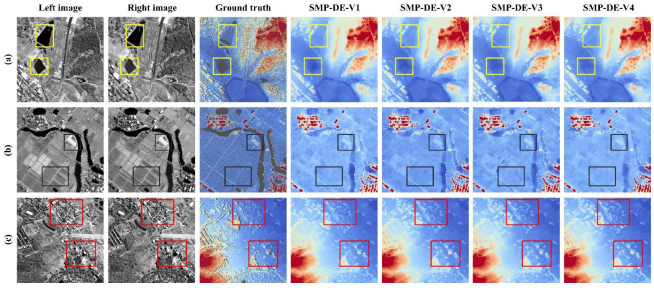


Fig. 10. Visualization examples of disparity estimation results from ablation experiments based on the WHU_Stereo dataset.

cost volume construction, cost aggregation, and fusion, and disparity calculation and optimization.

1) *SMP-DE-V2:* Upon the SMP-DE-V1, this variant incorporates the SMP-DED strategy.
2) *SMP-DE-V3:* Building on SMP-DE-V2, this variant further integrates the SMP-FCR module.
3) *SMP-DE-V4:* The MCUC module was added to SMP-DE-V3. The SMP-DE-V4 forms the complete SMP-DE network.

The specific configurations of multiple SMP-DE variants and their quantitative evaluation results based on the WHU_Stereo dataset are presented in Table V. All technical components significantly enhanced the accuracy of SMP-DE. Specifically, SMP-DE-V2, which incorporates the SMP-DED strategy, improved the disparity estimation quality for both the original test set and the extended dataset. As shown in Fig. 10, SMP-DE-V2 significantly reduced the number of disparity prediction outliers. This improvement can be attributed to the effective combination of the designed coarse-to-fine fault-tolerance training strategy, which minimizes the learning from erroneous data during training. As shown in the yellow-framed area of Fig. 10, SMP-DE-V3 enhanced with a sparse matching point-guided contrastive registration module demonstrates a significant improvement in disparity prediction smoothness in texture-sparse areas. Because of the MCUC module, SMP-DE-V4 notably ameliorated the issues of blurred and incorrect disparity estimations in texture-repetitive areas, as observed in the black-framed area in Fig. 10. Both SMP-DE-V3 and SMP-DE-V4 reduced erroneous disparity estimations in areas with shadow or lighting changes (red-framed area in Fig. 10) and optimized the disparity estimation accuracy for the extended test set. This proves that conducting network fitting across

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TABLE V

QUANTITATIVE EVALUATION RESULTS OF MULTIPLE NETWORK VARIANTS BASED ON THE WHU_STEREO DATASET

Network	Backbone	SMP-DED	SMP-CR	MCUC	Qichun		5 cities	
					D1	EPE	D1	EPE
SMP-DE-V1	√				9.60%	1.4983	14.42%	2.2774
SMP-DE-V2	√	√			9.32%	1.3829	13.75%	2.1205
SMP-DE-V3	√	√	√		9.39%	1.3808	11.81%	1.8758
SMP-DE-V4	√	√	√	√	9.04%	1.3633	11.59%	1.8463

multiple optimization spaces has a distinctly positive effect on the generalization performance of SMP-DE.

In this ablation study, the disparity estimation accuracy and generalization of the SMP-DE were progressively enhanced by incrementally adding meticulously designed fault-tolerance training strategies and optimized technical components. The experimental results validate the positive effects of the various technical processes and components designed in this study on SMP-DE.

B. Universality of the Method

In this section, further experiments were conducted using the proposed SMP-DED training strategy. The experiments involved combining other networks with the coarse-to-fine fault-tolerance training SMP-DED strategy. The purpose of the experiments was to demonstrate the effectiveness of the SMP-DED training strategy in assisting in avoiding erroneous data. To demonstrate the wide applicability of SMP-DED, the experiments employed a variety of networks and datasets. Various networks include FADNet based on a 2-D architecture, STTR based on a transformer architecture, PSMNet based on a 3-D architecture, and HMSM-Net designed for satellite stereo imagery. Experiments were conducted on the WHU_Stereo and US3D datasets.

Table VI presents the results of the generalizability assessment of the SMP-DED training strategy based on the WHU_Stereo dataset. All networks showed a notable improvement in the $D1$ metric on both the original and extended test sets after incorporating SMP-DED. Furthermore, the EPE accuracy of most methods improved after integrating SMP-DED, although only a few methods exhibited a slight decrease. Table VII presents the accuracy assessment results of the various end-to-end methods combined with the SMP-DED training strategy for the US3D dataset. The results indicate that all networks benefited from the fault-tolerance training strategy. Except for a minimal accuracy decrease in $D1$ for STTR in the Omaha test set, the $D1$ and EPE of the remaining methods showed improvements for both the original and extended test sets.

The experimental results obtained for both datasets validate that the proposed SMP-DED training strategy enhances the disparity estimation accuracy of most networks and is applicable to various satellite disparity estimation datasets.

C. Experiments on the Sufficient Training Set

In Section IV, the emphasis was placed on testing the accuracy and generalization of various stereo disparity estimation

TABLE VI

QUANTITATIVE RESULTS OF THE ASSESSMENT OF UNIVERSALITY FOR SMP-DED BASED ON THE WHU_STEREO DATASET

Method	Qichun				Other 5 cities			
	D1	EPE	ΔD1	ΔEPE	D1	EPE	ΔD1	ΔEPE
FADNet	22.30%	2.6177	↑2.80%	↓0.3845	38.64%	5.7596	↑7.21%	↑0.8886
FADNet'	19.50%	3.0022			31.43%	4.8710		
STTR	20.66%	2.7454	↑5.80%	↓0.1788	32.74%	5.1205	↑4.51%	↓0.3296
STTR'	14.86%	2.9242			28.23%	5.4501		
PSMNet	22.13%	2.3345	↑7.98%	↑0.4875	29.31%	3.5252	↑11.67%	↑1.0751
PSMNet'	14.15%	1.8470			17.64%	2.4501		
HMSM-Net	10.74%	1.5984	↑1.73%	↑0.1985	13.95%	2.1429	↑0.47%	↓0.0819
HMSM-Net'	9.01%	1.3999			13.48%	2.2248		

* ' represents the network combined with the SMP-DED training strategy.

TABLE VII

QUANTITATIVE RESULTS OF THE ASSESSMENT OF UNIVERSALITY FOR SMP-DED BASED ON THE US3D DATASET

Method	Omaha				Jackson			
	D1	EPE	ΔD1	ΔEPE	D1	EPE	ΔD1	ΔEPE
FADNet	7.68%	1.3455	↑0.07%	↑0.0144	17.49%	2.1154	↑0.26%	↑0.0352
FADNet'	7.59%	1.3599			17.23%	2.0802		
STTR	4.60%	1.0987	↓0.04%	↑0.0235	12.93%	1.7982	↑0.28%	↑0.0003
STTR'	4.64%	1.0752			12.65%	1.7979		
PSMNet	7.95%	1.4042	↑1.60%	↑0.1748	15.27%	1.9215	↑2.55%	↑0.2820
PSMNet'	6.35%	1.2294			12.72%	1.6395		
HMSM-Net	4.50%	1.0229	↑0.64%	↑0.0834	11.26%	1.5088	↑2.01%	↑0.1579
HMSM-Net'	3.86%	0.9395			9.25%	1.3509		

TABLE VIII

QUANTITATIVE EVALUATION RESULTS OF VARIOUS DISPARITY ESTIMATION METHODS ON THE MERGED DATASET

Method	WHU_Stereo Train set images:1200 Test set images: 128		US3D Train set images:3000 Test set images: 445	
	D1	EPE	D1	EPE
SGBM	13.39%	5.1167	10.92%	2.8514
FADNet	20.75%	2.3162	15.87%	1.9846
STTR	20.26%	2.7679	7.01%	1.2614
PSMNet	15.30%	2.0024	9.16%	1.3729
HMSM-Net	13.25%	1.9269	6.44%	1.1464
SMP-DE	**8.34%**	**1.3410**	**5.89%**	**1.0780**

algorithms. Thus, the data were partitioned into two subsets based on geographical regions in the experiment, which may result in an insufficient data volume in the original training dataset. To avoid insufficient data leading to inadequate deep learning model training and limited generalization capability, this section merged the original data and extended data from the WHU_Stereo and US3D datasets to form relatively large datasets (as shown in Table VIII).

The combined dataset was used as the data benchmark for algorithm training and testing. As shown in Table VIII, when compared to SGBM, the majority of deep learning algorithms continue to have an advantage on the test set. Among the deep-learning methods, FADNet, based on the 2-D convolution architecture, achieved lower evaluation accuracy. STTR and PSMNet are better than FADNet, but not as good as HMSM-Net. The accuracy advantages of the SMP-DE suggested in this study are still noticeable on both datasets, with $D1$ and EPE presenting values of 8.34% and 1.3410 for WHU_Stereo, respectively, and values of 5.89% and 1.0780 for US3D, respectively.

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TABLE IX

QUANTITATIVE ASSESSMENT RESULTS FROM VARIOUS METHODS FOR THE WHU_STEREO DATASET

WHU_Stereo test set	Method	Qichun		Other 5 cities	
		D1	EPE	D1	EPE
$T < 1.5$ Original test set images: 77 Extended test set images: 518	FADNet	21.37%	2.3232	43.46%	7.1111
	STTR	20.37%	2.5632	36.30%	6.0494
	PSMNet	22.19%	2.2637	32.32%	3.9832
	HMSM-Net	9.57%	**1.4395**	17.46%	2.4433
	SMP-DE	**8.96%**	1.4571	**13.90%**	**2.1441**
$T < 2.0$ Original test set images: 122 Extended test set images: 623	FADNet	22.71%	2.4545	44.42%	7.2248
	STTR	21.09%	2.7458	36.92%	6.1984
	PSMNet	25.63%	2.5102	34.88%	4.2627
	HMSM-Net	11.12%	**1.5138**	18.98%	2.5514
	SMP-DE	**11.09%**	1.6749	**15.37%**	**2.2901**

TABLE X

QUANTITATIVE ASSESSMENT RESULTS FROM VARIOUS METHODS FOR THE US3D DATASET

US3D test set	Method	Omaha		Jackson	
		D1	EPE	D1	EPE
$T < 1.5$ Original test set images: 365 Extended test set images: 1663	FADNet	9.79%	1.5496	19.31%	2.2900
	STTR	5.97%	1.1859	14.79%	1.9473
	PSMNet	9.84%	1.5556	17.27%	2.0936
	HMSM-Net	6.04%	1.2060	13.35%	1.6975
	SMP-DE	**5.12%**	**1.1334**	**10.77%**	**1.5278**
$T < 2.0$ Original test set images: 509 Extended test set images: 1946	FADNet	10.54%	1.6292	20.15%	2.3585
	STTR	5.98%	**1.1818**	15.55%	1.9969
	PSMNet	10.29%	1.5940	18.15%	2.1597
	HMSM-Net	6.68%	1.3149	14.36%	1.7815
	SMP-DE	**5.67%**	1.2598	**11.65%**	**1.6134**

D. Experiments on the Extensive Test Set

The experiment in Section IV used a restricted volume of high-quality data from the test set ($T < 1.0$). In order to validate the effectiveness of methods on a large test set, this section compared deep-learning methods on the WHU_Stereo and US3D test sets with further increased evaluation score thresholds T ($T < 1.5$ and $T < 2.0$).

Tables IX and X show that there has been a notable increase in the volume of data undergoing testing. Furthermore, SMP-DE offers notable benefits in terms of disparity estimate accuracy across a variety of T-values when compared to other methods. It demonstrates that SMP-DE is robust, in addition to further showing that fitting quality and generalization of SPE-DE are unaffected by the coarse-to-fine fault-tolerant training strategy.

E. Generalization Test Across Sensor Types

To further test the generalization performance of disparity estimation algorithms under different types of sensor data, cross-sensor type validation was conducted in this experiment. Specifically, training was conducted using the WHU_Stereo and US3D training sets, and evaluations were performed on the US3D and WHU_Stereo test sets, respectively (as shown in Table VIII). In addition, the ISPRS Vaihingen 3-D [37] aerial imagery dataset was used to further conduct cross-sensor performance verification.

Tables XI and XII present the cross-sensor evaluation results based on the WHU_Stereo and US3D training sets, respectively. The results revealed that SGBM showed a little accuracy variance among different data, and achieved the best

TABLE XI

GENERALIZATION PERFORMANCE EVALUATION BASED ON WHU_STEREO TRAINING SET

Training Dataset	Method	US3D		ISPRS Vaihingen 3D	
		D1	EPE	D1	EPE
WHU_Stereo	SGBM	**10.92%**	2.8514	**21.28%**	8.4691
	FADNet	57.78%	5.7451	91.28%	25.2655
	STTR	32.00%	3.4393	94.29%	29.6010
	PSMNet	24.60%	3.0563	44.12%	23.1430
	HMSM-Net	18.31%	2.7575	39.29%	7.4145
	SMP-DE	15.28%	**2.5863**	40.31%	**6.7762**

TABLE XII

GENERALIZATION PERFORMANCE EVALUATION BASED ON US3D TRAINING SET

Training Dataset	Method	WHU_Stereo		ISPRS Vaihingen 3D	
		D1	EPE	D1	EPE
US3D	SGBM	**13.39%**	5.1167	**21.28%**	**8.4691**
	FADNet	76.16%	11.6958	96.13%	40.7163
	STTR	37.41%	7.1861	92.17%	32.7528
	PSMNet	29.42%	5.6359	54.75%	22.7792
	HMSM-Net	22.40%	3.2158	42.94%	8.6376
	SMP-DE	21.43%	**3.1737**	42.70%	10.0905

$D1$ on all testing sets. On cross-sensor data, all deep learning disparity estimation methods exhibited a decline in accuracy. Furthermore, deep learning algorithms trained on US3D and WHU_Stereo showed a noticeable degradation in accuracy on the ISPRS Vaihingen 3-D dataset due to the significant differences in resolution, spectral distribution, and disparity distribution.

FADNet produced large disparity inference mistakes on cross-sensor data since it only uses 2-D convolution to infer disparity values. STTR, based on the Transformer architecture, also lacked strong cross-domain extrapolation capabilities and thus presented a significant decrease in accuracy on other kinds of data. With their 3-D convolution architectures, PSMNet and HMSM-Net showed a lower decline in accuracy. SMP-DE presented lower EPE values for the majority of cross-sensor test sets, and it was the most accurate network among the deep learning disparity estimation methods. These findings revealed the excellent generalization performance of the SMP-DE proposed in this article.

VI. CONCLUSION

The purpose of this study was to address problems caused by complex stereo-imaging areas and inaccuracies in the training data in the field of SSIDE. We proposed an innovative SMP-DE end-to-end universal framework. In the presence of errors in current SSIDE datasets and challenging imaging areas, SMP-DE achieves high-precision disparity estimation for satellite remote sensing images through two key aspects as follows.

First, in terms of network training, SMP-DE employs a SMP-DED training strategy. The solution involves precise quality assessment and encoding of training data pairs, as well as coarse-to-fine fault-tolerant adaptive training. Using this strategy, SMP-DE conducts refined network training while ensuring a sufficiently diverse training dataset. Therefore, the network guarantees rational fitting and accurate inference. Second, in terms of network training optimization, SMP-DE introduces the SMP-FCR and MCUC modules. These

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modules assist the network in homogeneity feature mining and heterogeneity feature capture within the feature representation space and regulate the distribution of the network cost volume in the matching cost space. Ultimately, SMP-DE undertakes comprehensive optimization across multiple spaces, effectively avoiding the occurrence of disparity-prediction noise and instability in challenging satellite imagery areas.

The experimental evidence demonstrates that SMP-DE outperforms other advanced methods in terms of disparity estimation accuracy and generalization performance. The ablation experiments described in the discussion section validate the functional effectiveness of the technical processes and components within SMP-DE. Furthermore, generalizability experiments effectively prove the transferability and universality of the SMP-DED training strategy across various methods and multiple datasets.

In the future, we will continue to focus on end-to-end disparity estimation and explore domain adaptation technologies. Adapting end-to-end disparity estimation network more effectively to a diverse range of satellite imagery, and providing more accurate disparity estimation.

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